Ex. No: 4

Analyze Email Headers and Detect Email Spoofing Using MHA

Date: 06-08-2026

Aim

To analyze the raw header of a received email using Microsoft Message Header Analyzer (MHA) and MXToolbox Email Header Analyzer, trace the email delivery path, verify SPF, DKIM, and DMARC authentication results, perform a WHOIS lookup on the originating IP address, and determine whether the email contains any signs of spoofing.

Description

Email header analysis is the process of examining the metadata attached to an email message to verify its authenticity and trace its delivery route. Important header fields such as **From**, **To**, **Return-Path**, **Received**, **Message-ID**, and authentication results (**SPF**, **DKIM**, and **DMARC**) help determine whether an email genuinely originated from the claimed sender or domain.

Tools such as **Microsoft Message Header Analyzer (MHA)** and **MXToolbox Email Header Analyzer** automatically decode complex email headers into a readable format, while **WHOIS** lookup services verify the ownership and location of the originating IP address.

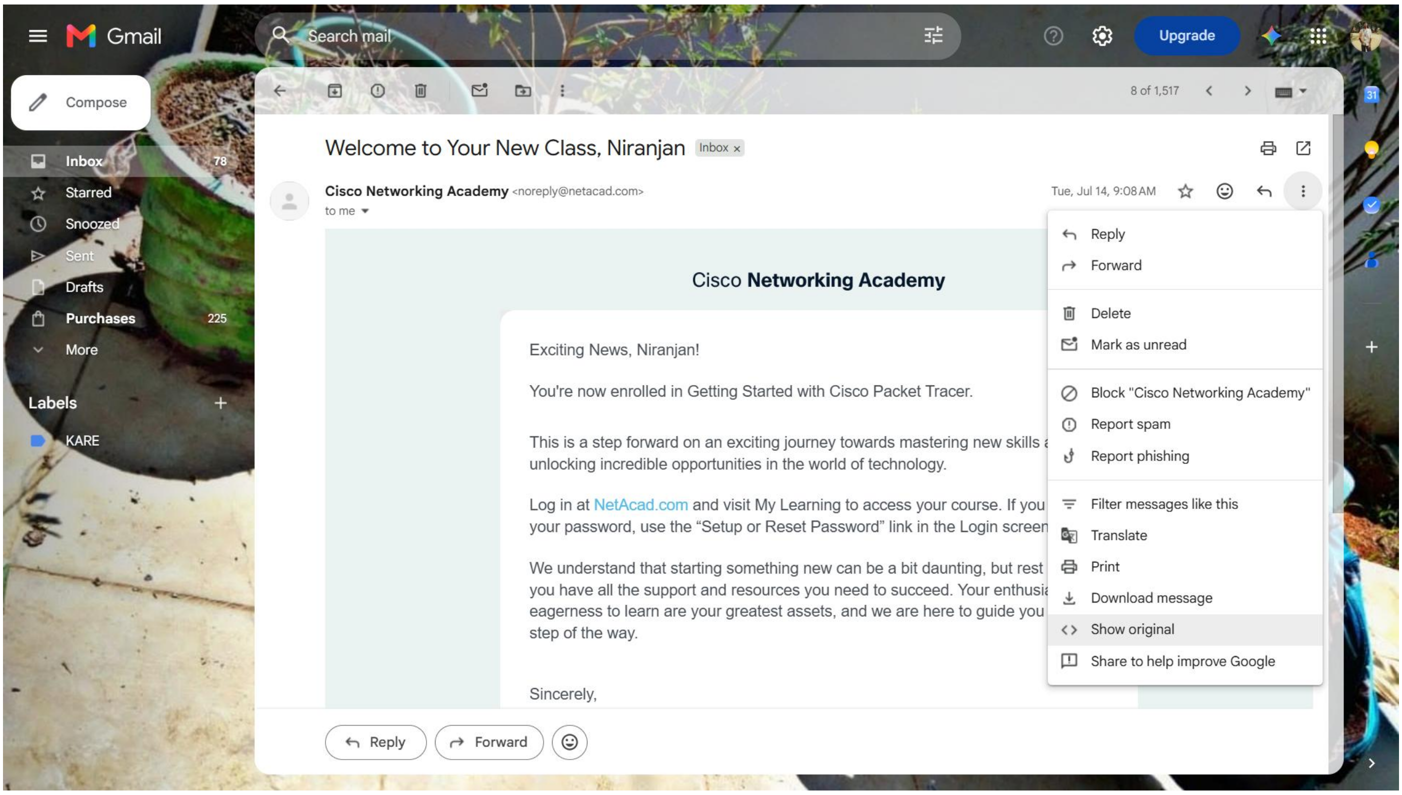
Procedure

1. Open the received email.
2. Extract the raw email header.
    - **Gmail:** More → **Show Original**
3. Copy the complete raw header (excluding the email body).
4. Identify important header fields:
    - From
    - To
    - Date
    - Subject
    - Return-Path
    - Received
    - Message-ID
5. Analyze the **Received** headers from **bottom to top** to trace the delivery path.
6. Perform a **WHOIS** lookup on the originating IP address.

7. Verify **SPF**, **DKIM**, and **DMARC** authentication results.
8. Compare the **Message-ID**, **Return-Path**, and **From** domains.
9. Look for anomalies such as:
    - Domain mismatches
    - Suspicious IP addresses
    - Abnormal timestamps
10. Analyze the header using:
    - Microsoft MHA
    - MXToolbox
11. Record the observations and conclude whether the email is legitimate or spoofed.

Sample Email Details

Field	**Value**
Subject	Welcome to Your New Class, Niranjan
From	Cisco Networking Academy <noreply@netacad.com>
To	thisisniranjan08@gmail.com
Date	Tue, 14 Jul 2026 03:38:04 +0000
Return-Path	<0100019f5eb405f6-...-000000@amazonses.com>
Message-ID	<0100019f5eb405f6-...-000000@email.amazonses.com>



Original Message

Message ID	<0100019f5eb405f6-7e46c55a-3534-4f17-a069-8416aa7a4427-000000@email.amazonses.com>
Created at:	Tue, Jul 14, 2026 at 9:08 AM (Delivered after 0 seconds)
From:	Cisco Networking Academy <noreply@netacad.com>
To:	thisisniranjan08@gmail.com
Subject:	Welcome to Your New Class, Niranjan
SPF:	PASS with IP 54.240.48.177 Learn more
DKIM:	'PASS' with domain netacad.com Learn more
DMARC:	'PASS' Learn more

Download Original

Copy to clipboard


```
Delivered-To: thisisniranjan08@gmail.com
Received: by 2002:a05:6359:6383:b0:2a2:acdc:9f8 with SMTP id sg3csp28094rwb; Mon, 13 Jul 2026 20:38:04 -0700 (PDT)
X-Received: by 2002:a05:6214:5011:b0:8ec:6811:78d2 with SMTP id 6a1803df08f44-903ff453489mr126600856d6.13.1784000284581; Mon, 13 Jul 2026
20:38:04 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1784000284; cv=none; d=google.com; s=arc-20260327;
b=BcEPuvrUkWoT9+hEzNf/sPDfMYzVTV/bXbdz+b+ws2WFJ/KlyvRodYgfEBrbTLsPKJ fhV7R6cPWg+6HiP1Z1PceTOYdZeoWysZl572J7xxCY3mh+yzvuytbGXNhtH91j/NvuF5
YnxXZJb6la7XgmA6JTP5E6dPT0eEg7zaDBFVi2VvQcGzODWe2La0pZcC0hAlgJ8EAjsK SaUQcJY+X510LKTzqDXr4YVN20KtbqmfKCSEx2Hl7n/08KiabxL5wbwVA/+eKtHmvQQ+
gc7OpN6O4fE2Uh5vxZeVKXKtVfY5YyMll69zyIoKFi1qIdN77QLKcS5FpHVa0RSN1ozd pYPA==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=feedback-id:mime-version:subject:message-
id:to:from:date :dkim-signature:dkim-signature; bh=LWxbUyOYxq27AqOrtfHhQMNM+P6zohTqn17Q4mPcHgU=;
fh=N50w4O4K9tBJrXIXLVS521paqRlopErg91gEVgXkiEs=; b=S5+ecWX2VgiCozIkrZBCAPvfXXmjFP3lvFgkfdYQOm51dnKicRda1meKxnRChgZ6Gb
QVdPxTvQaPalJVuJvsqWhqIVAl4yycFxURoJY+h+B3AWrLb8J9Vmw58JuDD+RsW3wB0R D0cvhkmIsfzzQaW02MCTDNhKIn/nEisZvhBKuQudTkcDGejJIMj0LTJLKOTG6DL4p4A6
```


Step 4: Received Header / Relay Path Analysis (MXToolbox)

MX TOOLBOX® SUPERTOOL

Pricing Tools Delivery Center Monitoring Products Resources Login

SuperTool MX Lookup Blacklists DMARC Diagnostics Email Health DNS Lookup **Analyze Headers** All Tools

Header Analyzed

Email Subject: Welcome to Your New Class, Niranjan

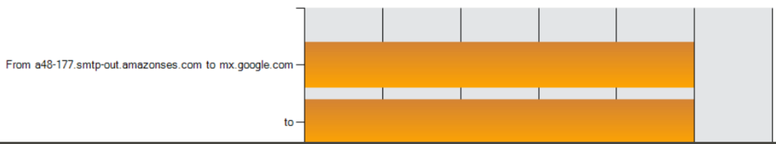
Analyze New Header

Delivery Information

- DMARC Compliant
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

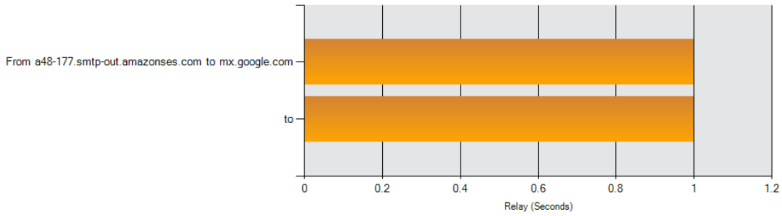
Relay Information

Received Delay:	**0 seconds**



Relay Information

Received Delay:	**0 seconds**



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	a48-177.smtp-out.amazonses.com 54.240.48.177	mx.google.com	ESMTPS	7/14/2026 3:38:04 AM	
2	0 seconds		2002:a05:6359:6383:b0:2a2:acdc:9f8	SMTP	7/14/2026 3:38:04 AM	

SPF and DKIM Information

dmarc:netacad.com

Show Solve Email Delivery Problems

v=DMARC1; p=reject; rua=mailto:fc92c7c69de4119@rep.dmarcanalyzer.com; ruf=mailto:fc92c7c69de4119@for.dmarcanalyzer.com; fo=1;

dkim:netacad.com:p74v6vscrr5r4ayt74lg7qugkee6oujj

Show

Dkim Public Record:

p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAtX0OkWpdPBtYW9Blb4npfv0V3wb3k4u1PKQ/ugHgG4a/MHirXG3DvqyQfenZY/0r1KKoh+WOxWdkz3dU6kNnZA5NiD+4tW3GeiBt7ZgddZZBtB

Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=p74v6vscrr5r4ayt74lg7qugkee6oujj; d=netacad.com; t=1784000284; h=Date:From:To:Message-ID:Subject:MIME-Vers

dkim:amazonses.com:224i4yxa5dv7c2xz3womw6peuasteono

Show

Dkim Public Record:

p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQC6LCYa2iGRMox5TLTaiN1tfg6dj09kuAbZWZOlHiAZoraUiiNMRZDmb00w+lffJ8dmbGApdp9FoApoKuEk6MH87NPyyL5otrq6wk1ozAZ12cQZDlBiS0

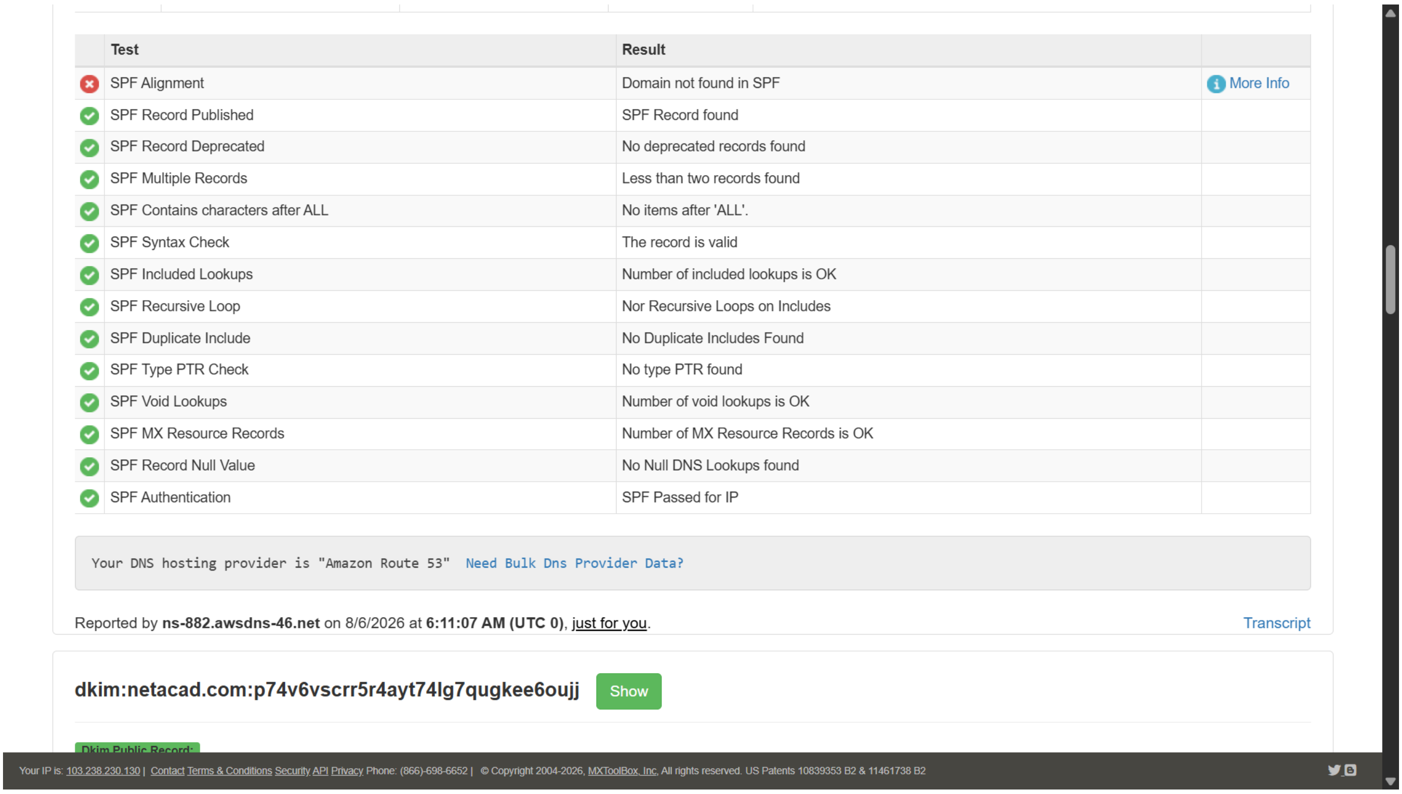
Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=224i4yxa5dv7c2xz3womw6peuasteono; d=amazonses.com; t=1784000284; h=Date:From:To:Message-ID:Subject:MIME-Ve

Headers Found

Header Name	Header Value

Your IP is: 103.238.230.130 | Contact Terms & Conditions Security API Privacy Phone: (866)-698-6652 | © Copyright 2004-2026, MXToolBox, Inc, All rights reserved. US Patents 10839353 B2 & 11461738 B2



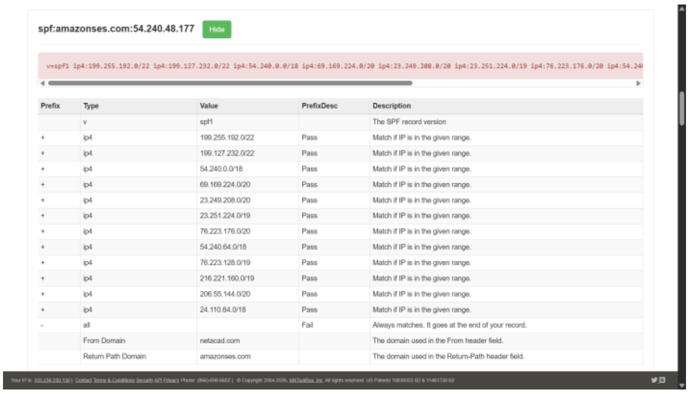
	Test	Result	
✖	SPF Alignment	Domain not found in SPF	More Info
✔	SPF Record Published	SPF Record found	
✔	SPF Record Deprecated	No deprecated records found	
✔	SPF Multiple Records	Less than two records found	
✔	SPF Contains characters after ALL	No items after 'ALL'.	
✔	SPF Syntax Check	The record is valid	
✔	SPF Included Lookups	Number of included lookups is OK	
✔	SPF Recursive Loop	Nor Recursive Loops on Includes	
✔	SPF Duplicate Include	No Duplicate Includes Found	
✔	SPF Type PTR Check	No type PTR found	
✔	SPF Void Lookups	Number of void lookups is OK	
✔	SPF MX Resource Records	Number of MX Resource Records is OK	
✔	SPF Record Null Value	No Null DNS Lookups found	
✔	SPF Authentication	SPF Passed for IP	

```
Your DNS hosting provider is "Amazon Route 53"  Need Bulk Dns Provider Data?
```

Reported by **ns-882.awsdns-46.net** on 8/6/2026 at **6:11:07 AM (UTC 0)**, just for you. Transcript

dkim:netacad.com:p74v6vscrr5r4ayt74lg7qugkee6oujj Show

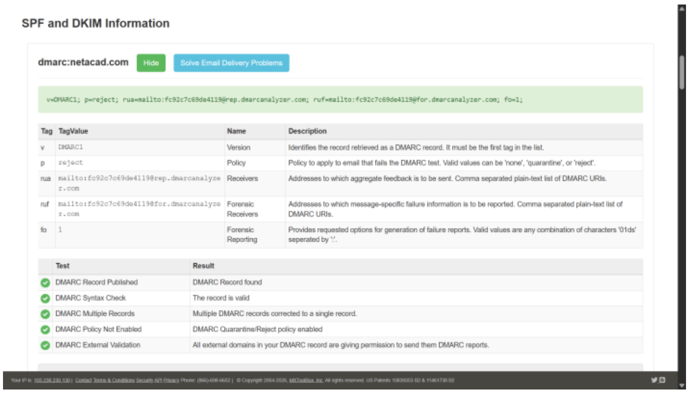
Dkim Public Record:



spf:amazonses.com:54.240.48.177 Hide

```
v=spf1 ip4:199.255.192.0/22 ip4:199.127.232.0/22 ip4:54.240.0.0/18 ip4:69.169.224.0/20 ip4:23.249.208.0/20 ip4:23.251.224.0/19 ip4:76.223.176.0/20 ip4:54.24
```

Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
+	ip4	199.255.192.0/22	Pass	Match if IP is in the given range.
+	ip4	199.127.232.0/22	Pass	Match if IP is in the given range.
+	ip4	54.240.0.0/18	Pass	Match if IP is in the given range.
+	ip4	69.169.224.0/20	Pass	Match if IP is in the given range.
+	ip4	23.249.208.0/20	Pass	Match if IP is in the given range.
+	ip4	23.251.224.0/19	Pass	Match if IP is in the given range.
+	ip4	76.223.176.0/20	Pass	Match if IP is in the given range.
+	ip4	54.240.64.0/18	Pass	Match if IP is in the given range.
+	ip4	76.223.128.0/19	Pass	Match if IP is in the given range.
+	ip4	216.221.160.0/19	Pass	Match if IP is in the given range.
+	ip4	206.55.144.0/20	Pass	Match if IP is in the given range.
+	ip4	24.110.64.0/18	Pass	Match if IP is in the given range.
-	all		Fail	Always matches. It goes at the end of your record.
	From Domain	netacad.com		The domain used in the From header field.
	Return Path Domain	amazonses.com		The domain used in the Return-Path header field.



SPF and DKIM Information

dmarc:netacad.com Hide Solve Email Delivery Problems

```
v=DMARC1; p=reject; rua=mailto:fc92c7c69de4119@rep.dmarcanalyzer.com; ruf=mailto:fc92c7c69de4119@for.dmarcanalyzer.com; fo=1;
```

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	reject	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
rua	mailto:fc92c7c69de4119@rep.dmarcanalyzer.com	Receivers	Addresses to which aggregate feedback is to be sent. Comma separated plain-text list of DMARC URIs.
ruf	mailto:fc92c7c69de4119@for.dmarcanalyzer.com	Forensic Receivers	Addresses to which message-specific failure information is to be reported. Comma separated plain-text list of DMARC URIs.
fo	1	Forensic Reporting	Provides requested options for generation of failure reports. Valid values are any combination of characters '01ds' seperated by ':'.

	Test	Result
✔	DMARC Record Published	DMARC Record found
✔	DMARC Syntax Check	The record is valid
✔	DMARC Multiple Records	Multiple DMARC records corrected to a single record.
✔	DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled
✔	DMARC External Validation	All external domains in your DMARC record are giving permission to send them DMARC reports.

Hop	Delay	From	By	Protocol	Time (UTC)
1	—	a48-177.smtp-out.amazonses.com (54.240.48.177)	mx.google.com	ESMTPS	14-07-2026 03:38:04
2	0 sec	2002:a05:6359:6383:...	Google Internal	SMTP	14-07-2026 03:38:04

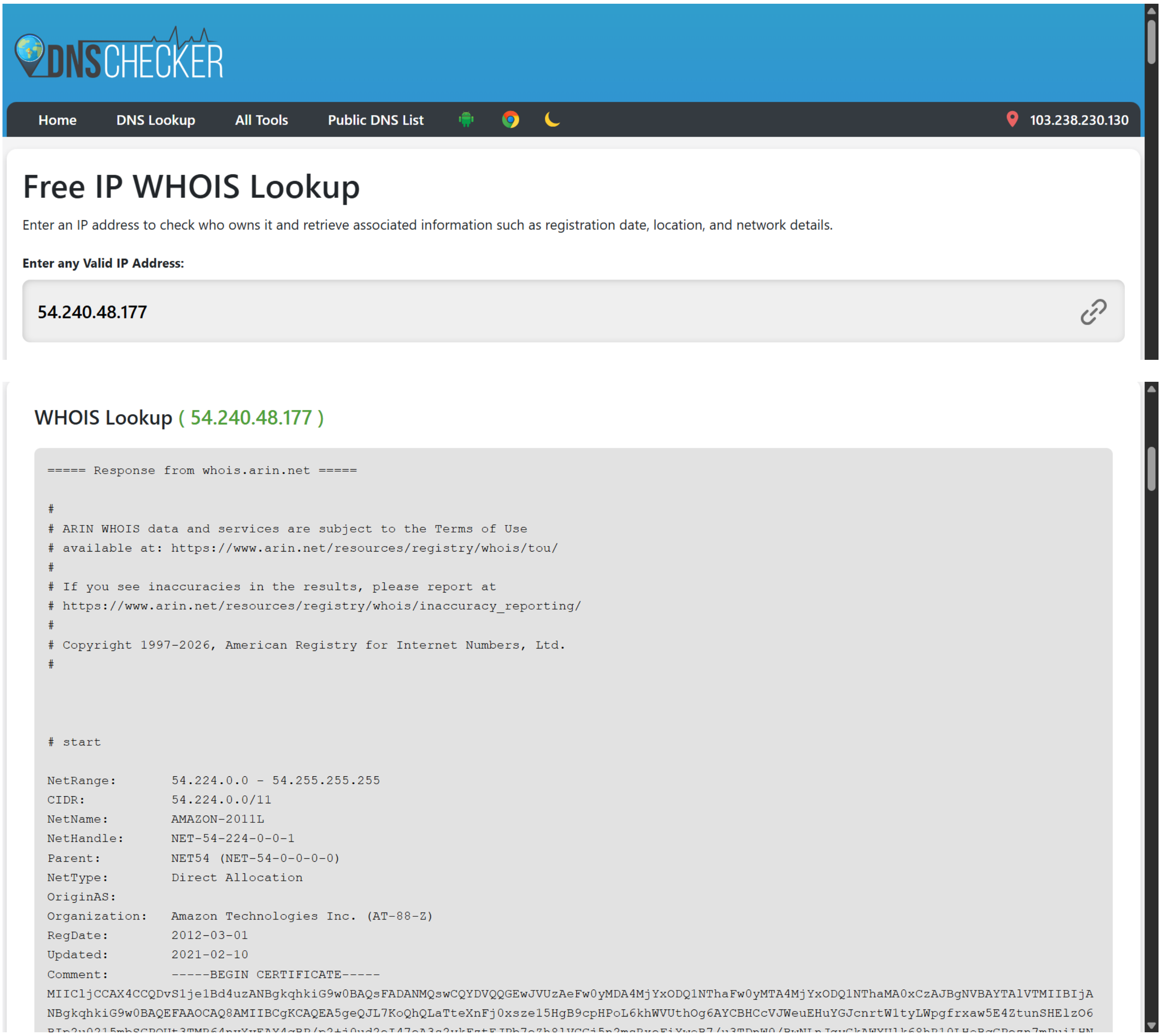
Observation

- The email passed through only **two relay hops**.
- No abnormal delay or unexpected mail server was observed.
- The delivery path appears normal and legitimate.

Step 5: WHOIS Lookup of Originating IP Address

Originating IP: 54.240.48.177

Field	Value
NetRange	54.240.0.0 – 54.240.63.255
CIDR	54.240.0.0/18
NetName	AWSEMAIL-Z
Organization	Amazon Web Services, Inc.
Parent Block	Amazon Technologies Inc.
Location	Seattle, Washington, USA
Abuse Contact	Amazon SES Abuse (ARIN)



```
OrgRoutingHandle: IPROU3-ARIN
OrgRoutingName:   IP Routing
OrgRoutingPhone:  +1-206-555-0000
OrgRoutingEmail:  aws-routing-poc@amazon.com
OrgRoutingRef:    https://rdap.arin.net/registry/entity/IPROU3-ARIN

OrgTechHandle: ANO24-ARIN
OrgTechName:   Amazon EC2 Network Operations
OrgTechPhone:  +1-206-555-0000
OrgTechEmail:  amzn-noc-contact@amazon.com
OrgTechRef:    https://rdap.arin.net/registry/entity/ANO24-ARIN

# end



#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#
```

IP WHOIS - Check IP Address Registration and Ownership Details

```
# start

NetRange:       54.240.0.0 - 54.240.63.255
CIDR:           54.240.0.0/18
NetName:        AWSEMAIL-Z
NetHandle:      NET-54-240-0-0-2
Parent:         AMAZON-2011L (NET-54-224-0-0-1)
NetType:        Reallocated
OriginAS:
Organization:   Amazon Web Services, Inc. (AMAZO-22)
RegDate:        2012-01-03
Updated:        2021-02-10
Ref:            https://rdap.arin.net/registry/ip/54.240.0.0


OrgName:        Amazon Web Services, Inc.
OrgId:          AMAZO-22
Address:        Amazon Web Services
Address:        1200 12th Avenue South
City:           Seattle
StateProv:      WA
PostalCode:     98144
Country:        US
RegDate:        2010-03-08
Updated:        2026-04-17
Ref:            https://rdap.arin.net/registry/entity/AMAZO-22


OrgNOCHandle: AANO1-ARIN
OrgNOCName:   Amazon AWS Network Operations
OrgNOCPhone:  +1-206-555-0000
OrgNOCEmail:  amzn-noc-contact@amazon.com
OrgNOCRef:    https://rdap.arin.net/registry/entity/AANO1-ARIN
```

```
OrgName:        Amazon Technologies Inc.
OrgId:          AT-88-Z
Address:        410 Terry Ave N.
City:           Seattle
StateProv:      WA
PostalCode:     98109
Country:        US
RegDate:        2011-12-08
Updated:        2026-04-17
Comment:        All abuse reports MUST include:
Comment:        * src IP
Comment:        * dest IP (your IP)
Comment:        * dest port
Comment:        * Accurate date/timestamp and timezone of activity
Comment:        * Intensity/frequency (short log extracts)
Comment:        * Your contact details (phone and email) Without these we will be unable to identify the correct owner of the IP
address at that point in time.
Ref:            https://rdap.arin.net/registry/entity/AT-88-Z


OrgNOCHandle: AANO1-ARIN
OrgNOCName:   Amazon AWS Network Operations
OrgNOCPhone:  +1-206-555-0000
OrgNOCEmail:  amzn-noc-contact@amazon.com
OrgNOCRef:    https://rdap.arin.net/registry/entity/AANO1-ARIN

OrgRoutingHandle: ARMP-ARIN
OrgRoutingName:   AWS RPKI Management POC
OrgRoutingPhone:  +1-206-555-0000
OrgRoutingEmail:  aws-rpki-routing-poc@amazon.com
OrgRoutingRef:    https://rdap.arin.net/registry/entity/ARMP-ARIN

OrgTechHandle: ANO24-ARIN
```

```
# start

NetRange:       54.224.0.0 - 54.255.255.255
CIDR:           54.224.0.0/11
NetName:        AMAZON-2011L
NetHandle:      NET-54-224-0-0-1
Parent:         NET54 (NET-54-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Amazon Technologies Inc. (AT-88-Z)
RegDate:        2012-03-01
Updated:        2021-02-10
Comment:        -----BEGIN CERTIFICATE-----
```

MIICljCCAX4CCQDvS1je1Bd4uzANBgkqhkiG9w0BAQsFADANMQswCQYDVQQGEwJVUzAeFw0yMDA4MjYxODQ1NThaFw0yMTA4MjYxODQ1NThaMA0xCzAJBgNVBAYTAlVTMIIBIjA
NBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA5geQJL7KoQhQLaTteXnFj0xsze15HgB9cpHPoL6khWVUthOg6AYCBHCcVJWeuEHuYGJcnrtW1tyLWpgfrxaw5E4ZtunSHElzO6
BIp2u0215mbSGPQUt3TMR64nvXvEAY4qBP/p2+j0ud2eI47eA3s2ykFztEJPb7eZh8lVCGj5n2msRxeFiYwoB7/u3TDnW0/BwNLnJgyGkAWYUlk68hR10LHoBqGPezn7mPuiLHN
a6JQP0WTYBz/80kS3m/4oZ7NS20PMieXqFjfYEgW6fPg7uJKhH3aYVVveZpBS5cRzm360HyT5hj1rUJh34nVCLMlvP+400w1wxr9buLnQzVlwIDAQABMA0GCSqGSIb3DQEBCwUA
A4IBAQCZD7ERFb2LpeLdQgyji/ZqZ7lDXR8wq4m+ihMiqpPcwTVs1dfBfKDvZ4K6Ddyzkfd1NQYPWiV47nvqgJxwdISa7vN011RxBEGkYdJ8cNaRXW7aCGfQ8ZSQL6mbXsm4sbv
DQNHiWJcdUB0KTzR/wpbXf9+24TbPGaOsZvfnKtdllZhY5xFiOVCOdI59c/XyDH9aqOKNE0pOeATX55I3bU5PKeK5CM8oAtD2sFAQ956Uvj7/vFDs8QP3upzf53R+erSU10L1fT
QBWHjNUCcf9wviS+U4hsaCcBZMlw6d5Q84GYX1tS+YwtA0Fv/NQcOWr9RJT+JVnpbyAxEyjI37XOqH-----END CERTIFICATE-----

```
Ref:            https://rdap.arin.net/registry/ip/54.224.0.0



OrgName:        Amazon Technologies Inc.
OrgId:          AT-88-Z
Address:        410 Terry Ave N.
City:           Seattle
StateProv:      WA
PostalCode:     98109
Country:        US
RegDate:        2011-12-08
Updated:        2026-04-17
```

Observation

The IP address belongs to **Amazon Web Services (AWS)** under the **Amazon Simple Email Service (SES)** network. This confirms that the email originated from a legitimate Amazon SES mail server.

Step 6: SPF, DKIM and DMARC Authentication

Authentication	**Result**	**Details**
SPF	✅ Pass	Amazon SES authorizes IP **54.240.48.177** to send email.
DKIM (netacad.com)	✅ Pass	Valid digital signature using **d=netacad.com**.
DKIM (amazonses.com)	✅ Pass	Amazon SES signature verified successfully.
DMARC	✅ Pass	Domain alignment successful (**header.from = netacad.com**).

Step 7 & 8: Message-ID and Anomaly Analysis

Domain Comparison

Header Field	**Domain**
From	netacad.com
Return-Path	amazonses.com
Message-ID	email.amazonses.com

Analysis

Although the Return-Path and Message-ID belong to Amazon SES, the visible sender domain is netacad.com.

This is not considered spoofing because Cisco Networking Academy legitimately uses Amazon Simple Email Service (SES) as its email delivery provider.

The DKIM signature is correctly aligned with netacad.com, allowing DMARC authentication to pass successfully.

Therefore, the domain mismatch is a normal characteristic of third-party email service providers and not evidence of spoofing.

Step 9: Tools Used

Tool	Purpose
Microsoft Message Header Analyzer (MHA)	Parsed raw email headers into a readable format
MXToolbox Email Header Analyzer	Verified relay path and SPF/DKIM/DMARC authentication
WHOIS (ARIN)	Verified ownership and location of the originating IP

Results

- SPF Authentication: **Passed**
- DKIM Authentication: **Passed**
- DMARC Authentication: **Passed**
- Relay Path: **Normal**
- Originating IP: **Verified (Amazon SES)**
- Domain Alignment: **Valid**
- Spoofing Indicators: **None**

Conclusion

The analyzed email titled **"Welcome to Your New Class, Niranjan"** from **Cisco Networking Academy** successfully passed **SPF**, **DKIM**, and **DMARC** authentication checks.

The **Received** headers show a legitimate two-hop email delivery path with no unusual delays or routing anomalies. A WHOIS lookup confirmed that the originating IP address (**54.240.48.177**) belongs to **Amazon Web Services (AWS)** and is part of the **Amazon Simple Email Service (SES)** infrastructure.

Although the **Return-Path** and **Message-ID** domains differ from the visible **From** domain, this is expected because Cisco Networking Academy uses Amazon SES as its authorized email service provider. The valid DKIM signature ensures proper DMARC alignment, confirming the email's authenticity.

Final Verdict

Parameter	Status
Email Authenticity	Legitimate
Email Spoofing Detected	No
Overall Verdict	**LEGITIMATE EMAIL**